

Question 2

$$(\Delta r)(r \Delta \theta)(\Delta z) = dV$$

Two radial faces at r and $r + \Delta r$

$$A_r = r \Delta \theta \Delta z$$

$$\text{Mass flow in: } (\rho v_r)_r (r \Delta \theta \Delta z)$$

$$\text{Mass flow out: } (\rho v_r)_{r+\Delta r} ((r+\Delta r) \Delta \theta \Delta z)$$

$$\begin{aligned} \text{Net: } & [(\rho v_r)(r+\Delta r) \Delta \theta \Delta z]_{r+\Delta r} - [(\rho v_r) r \Delta \theta \Delta z]_r \\ & = \frac{\partial}{\partial r} (r \rho v_r) \Delta r \Delta \theta \Delta z \end{aligned}$$

In θ direction

$$A_\theta = \Delta r \Delta z$$

$$\text{Mass in: } (\rho v_\theta)_\theta (\Delta r \Delta z)$$

$$\text{Mass out: } (\rho v_\theta)_{\theta+\Delta\theta} (\Delta r \Delta z)$$

$$\begin{aligned} \text{Net: } & [(\rho v_\theta)_{\theta+\Delta\theta} - (\rho v_\theta)_\theta] \Delta r \Delta z \\ & = \frac{\partial \rho v_\theta}{\partial \theta} \Delta \theta \Delta r \Delta z \end{aligned}$$

In z direction

$$A_z = r \Delta r \Delta \theta$$

$$\text{Mass in: } (\rho v_z)_z (r \Delta r \Delta \theta)$$

$$\text{Mass out: } (\rho v_z)_{z+\Delta z} (r \Delta r \Delta \theta)$$

$$\begin{aligned} \text{Net: } & [(\rho v_z)_{z+\Delta z} - (\rho v_z)_z] r \Delta r \Delta \theta \\ & = \frac{\partial \rho v_z}{\partial z} \Delta z r \Delta r \Delta \theta \end{aligned}$$

$$\text{Accumulation: mass inside} \rightarrow \rho(r \Delta r \Delta \theta \Delta z)$$

$$= \frac{\partial \rho}{\partial t} (r \Delta r \Delta \theta \Delta z)$$

Continuity $\rightarrow$ Accumulation + net outflow = 0

$$\rightarrow \frac{\partial \rho}{\partial t} (r \Delta r \Delta \theta \Delta z) + \frac{\partial}{\partial r} (\rho v_r) \Delta r \Delta \theta \Delta z$$

$$+ \frac{\partial (\rho v_\theta)}{\partial \theta} \Delta r \Delta \theta \Delta z + \frac{\partial (\rho v_z)}{\partial z} r \Delta r \Delta \theta \Delta z = 0$$

Divide by $r \Delta r \Delta \theta \Delta z$

$$\rightarrow \boxed{\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial}{\partial r} (r \rho v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\rho v_\theta) + \frac{\partial}{\partial z} (\rho v_z) = 0}$$

Problem 3A.2

$$P = T_z \omega_i = 4 \pi \mu \omega_i^2 R^2 L \left(\frac{\mu^2}{1 - \mu^2} \right)$$

20 bearings, 1 bearing = 1 ft

$$L = 2 \cdot 20 \cdot 1 = 40 \text{ ft} \xrightarrow{\text{to m}} 40 \cdot \frac{12}{39.37} = 12.2 \text{ m}$$

$$d = 16 \text{ in} \cdot \frac{1}{2} = 8 \text{ in} = R \xrightarrow{\text{to m}} 8 / 39.37 = (0.203 \text{ m})^2 = 0.04129 \text{ m}^2$$

$$\text{Viscosity} = 5000 \text{ cp} \cdot 10^{-3} \frac{\text{kg}}{\text{m} \cdot \text{s}} = 5 \text{ kg/m} \cdot \text{s} = \mu$$

$$\omega_i = \left(\frac{50}{60} \right) (2\pi) = \frac{5\pi}{3} \text{ rad/s}$$

$$0.005 \text{ in Clearance } R_2 = 8 + 0.005 = 8.005 \text{ in}$$

$$\mu = \frac{R_1}{R_2} = \frac{8}{8.005} = 0.999375$$

$$\frac{\mu^2}{1 - \mu^2} = \frac{(0.999375)^2}{1 - (0.999375)^2} = 799.6$$

$$P = 4\pi (5) \left(\frac{5\pi}{3} \right)^2 (0.04129) (12.2) (799.6) = 693832.46 \text{ W}$$

$$1 \text{ hp} = 746 \text{ W} \xrightarrow{\text{divide}} 930.07 \text{ hp}$$

$$\text{Two engines} \rightarrow 4000 \text{ hp} \cdot 2 = 8000 \text{ hp}$$

$$\rightarrow \frac{930.07 \text{ hp}}{8000 \text{ hp}} = \boxed{0.116 \text{ fraction of power expended}}$$

Problem 3B.1

a) $V_\theta = V_\theta(r) \quad V_r = V_z = 0$

$$\int \frac{1}{r} \frac{d}{dr} \left(r \frac{d}{dr} \left(\frac{V_\theta}{r} \right) \right) dr = 0$$

$$\rightarrow \int \frac{d}{dr} \left(\frac{V_\theta}{r} \right) dr = \int \frac{C_1}{r} dr \rightarrow \frac{V_\theta}{r} = C_1 + C_2 \left(\frac{R}{r} \right)^2$$

$$V_\theta(1R) = -\omega_1(1R) \quad V_\theta(R) = -\omega_0 R$$

$$\frac{V_\theta}{r} = -\omega$$

$$-\omega_1 = D_1 + D_2 \frac{1}{1R^2} \quad -\omega_0 = D_1 + D_2$$

$$-\omega_1 - (-\omega_0) = D_2 \left(\frac{1}{1R^2} - 1 \right) \rightarrow D_2 = \frac{1R^2(-\omega_0 + \omega_1)}{1R^2 - 1}$$

$$\text{Substitute} \rightarrow D_1 = \frac{-\omega_1 1R^2 - \omega_0}{1R^2 - 1}$$

$$\frac{V_\theta}{r} = \frac{-\omega_0 - \omega_1 1R^2}{1 - 1R^2} + \frac{-\omega_1 - \omega_0}{1 - 1R^2} \left(\frac{1R}{r} \right)^2 \quad | \cdot r$$

$$V_\theta = \frac{1R}{1 - 1R^2} \left[(-\omega_0 - \omega_1 1R^2) \left(\frac{r}{1R} \right) + (-\omega_1 - \omega_0) \left(\frac{1R}{r} \right) \right]$$

For torque $\rightarrow \tau_{r\theta} = -Mr \frac{d}{dr} \left(\frac{V_\theta}{r} \right)$

outer $T_z = \int_0^L \int_0^{2\pi} (-\tau_{r\theta} R) R d\theta dz$

$$T_z = 2\pi L R^2 \left[Mr \frac{d}{dr} \left(\frac{V_\theta}{r} \right) \right] = -4\pi M L (-\omega_1 - \omega_0) \frac{(1R)^2}{1 - 1R^2}$$

inner $T_z = +4\pi M L (-\omega_1 - \omega_0) \frac{(1R)^2}{1 - 1R^2}$

↑ opposite sign as they oppose each other

b) $V_\phi = V_\phi(r, \theta) \quad \frac{V_\phi}{r \sin \theta} = D_1 + D_2 \left(\frac{R}{r} \right)^3$

$$V_\phi(1R) = 1R \omega_1 \sin \theta \quad V_\phi(R) = R \omega_0 \sin \theta$$

Divide by $r \sin \theta \rightarrow -\omega_1 = D_1 + D_2 \frac{1}{1R^3} \quad -\omega_0 = D_1 + D_2$

$$D_1 = \frac{-\omega_0 - \omega_1 1R^3}{1 - 1R^3} \quad D_2 = \frac{1R^3(-\omega_1 + \omega_0)}{1 - 1R^3}$$

Substitute D_1 and D_2

$$\rightarrow V_\phi = \frac{\mu R}{1-\mu^2} \left[(\omega_0 - \omega_i \mu^2) \left(\frac{r}{\mu R} \right) + (\omega_i - \omega_0) \left(\frac{\mu R}{r} \right)^2 \right] \sin \theta$$

For torque

$$T_z = \int_0^{2\pi} \int_0^\pi (-\tau_{r\phi}) (R \sin \theta) R^2 \sin \theta d\theta d\phi$$

$$= \int_0^{2\pi} \int_0^\pi \left[\mu r \frac{d}{dr} \left(\frac{V_\phi}{r} \right) \right] (R \sin \theta) R^2 d\theta d\phi$$

$$\text{outer } T_z = -8\pi\mu(\omega_i - \omega_0) \frac{(\mu R)^3}{1-\mu^2}$$

$$\text{inner } T_z = \int_0^{2\pi} \int_0^\pi (+\tau_{r\phi}) (\mu R \sin \theta) (\mu R)^2 \sin \theta d\theta d\phi$$

$$= +8\pi\mu(\omega_i - \omega_0) \frac{(\mu R)^3}{1-\mu^2}$$

Problem 3B.4

$$V_\theta = V_\theta(r, \theta) \quad N_r = V_\phi = 0$$

$$a) \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 V_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (V_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial V_\phi}{\partial \phi} = 0$$

$$\text{Multiply both sides by } r \sin \theta \rightarrow \frac{\partial}{\partial \theta} (V_\theta \sin \theta) = 0$$

$$\text{can only depend on } r \rightarrow V_\theta \sin \theta = u(r)$$

$$b) \rho(\text{div } \mathbf{v}) \approx 0 \rightarrow 0 = -\nabla^2 P + \mu \nabla^2 V$$

$$\theta \text{ component in Bird} \rightarrow 0 = -\frac{1}{r} \frac{\partial P}{\partial \theta} + \mu \left[\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV_\theta}{dr} \right) \right]$$

$$\text{Substitute } \rightarrow V_\theta = \frac{u(r)}{\sin \theta} \rightarrow \frac{dV_\theta}{dr} = \frac{1}{\sin \theta} \frac{du}{dr}$$

$$\text{So } \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dV_\theta}{dr} \right) = \frac{1}{\sin \theta} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right)$$

$$\rightarrow 0 = -\frac{1}{r} \frac{\partial P}{\partial \theta} + \mu \left[\frac{1}{\sin \theta r^2} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) \right]$$

c) Multiply equation by $r \sin \theta$

$$0 = -\sin \theta \frac{dP}{d\theta} + \frac{M}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right)$$

first term function of θ , second term function of r

both equal to constant, B

$$\sin \theta \frac{dP}{d\theta} = B \quad \frac{M}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = B$$

$$\sin \theta \frac{dP}{d\theta} = B \rightarrow \frac{dP}{d\theta} = \frac{B}{\sin \theta}$$

$$\text{Integrate } \int_{P_1}^{P_2} dP = \int_{\epsilon}^{\pi-\epsilon} \frac{B}{\sin \theta} d\theta$$

$$\int \frac{d\theta}{\sin \theta} = \ln \tan \left(\frac{\theta}{2} \right) \text{ identity}$$

$$\rightarrow P_2 - P_1 = B \ln \left(\frac{\cot(\epsilon/2)}{\tan(\epsilon/2)} \right)$$

$$B = \frac{P_2 - P_1}{2 \ln(\cot(\epsilon/2))} \leftarrow \text{Pressure}$$

$$\text{Velocity} \rightarrow \frac{M}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = B \quad | \cdot \frac{r}{M}$$

$$\int \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = \int \frac{Br}{M}$$

$$r^2 \frac{du}{dr} = \frac{Br^2}{2M} + C_1 \quad | \cdot \frac{1}{r^2}$$

$$\int \frac{du}{dr} = \int \frac{Br}{2M} + \frac{C_1}{r^2}$$

$$u(r) = \frac{B}{2M} r - \frac{C_1}{r} + C_2$$

$$u(r=R) = 0, \quad u(r=R) = 0 \rightarrow C_1 = -k \quad C_2 = -(k/R)$$

$$\text{Substitute and combine} \rightarrow u(r) = \frac{(P_1 - P_2)R}{4M \ln(\cot(\epsilon/2))} \left[\left(1 - \frac{r}{R}\right) + k \left(1 - \frac{R}{r}\right) \right]$$

$$d) \text{ We will use } \theta = \frac{\pi}{2} \text{ b/c } v_\theta = \frac{u(r)}{\sin \theta} = u(r)$$

$$W = \int p v_\theta dA \quad dA = r dr d\phi$$

$$\rightarrow W = \int_0^{2\pi} \int_{MR}^R p u(r) r dr d\phi \quad \xi = \frac{r}{R}$$

$$W = \int_0^{2\pi} \int_{MR}^R p v_\theta \Big|_{\theta=\frac{\pi}{2}} r dr d\phi = 2\pi R^2 p \int_M^1 u \xi d\xi$$

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$$= 2\pi R^2 \rho \frac{(\rho_1 - \rho_2) R}{4\mu \ln(\cot(\frac{\epsilon}{2}))} \int_M \left[(1 - \epsilon) + M(1 - \frac{1}{3}) \right] \rho d\epsilon$$

$$= \boxed{2\pi R^2 \rho \frac{(\rho_1 - \rho_2) R}{4\mu \ln(\cot(\frac{\epsilon}{2}))} \cdot \frac{1}{6} (1 - M)^3}$$

#1

$$\frac{\partial \rho}{\partial t} = -\operatorname{div}(\rho \mathbf{v}) \rightarrow \frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{v}) = 0$$

$$\frac{\partial}{\partial t} \left(\rho \frac{v^2}{2} \right) + \operatorname{div} \left(\rho \frac{v^2}{2} \mathbf{v} \right) = \rho \frac{D}{Dt} \frac{v^2}{2}$$

$$\rightarrow \frac{\partial}{\partial t} \left(\rho \frac{v^2}{2} \right) = \frac{v^2}{2} \frac{\partial \rho}{\partial t} + \rho \frac{\partial}{\partial t} \left(\frac{v^2}{2} \right)$$

$$\operatorname{div} \left(\rho \frac{v^2}{2} \mathbf{v} \right) = \frac{v^2}{2} \operatorname{div}(\rho \mathbf{v}) + \rho \mathbf{v} \cdot \nabla \left(\frac{v^2}{2} \right)$$

$$\text{Add both} \rightarrow \frac{v^2}{2} \frac{\partial \rho}{\partial t} + \rho \frac{\partial}{\partial t} \left(\frac{v^2}{2} \right) + \frac{v^2}{2} \operatorname{div}(\rho \mathbf{v}) + \rho \mathbf{v} \cdot \nabla \left(\frac{v^2}{2} \right)$$

$$\frac{v^2}{2} \left[\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{v}) \right] + \rho \left[\frac{\partial}{\partial t} \left(\frac{v^2}{2} \right) + \mathbf{v} \cdot \nabla \left(\frac{v^2}{2} \right) \right]$$

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{v}) = 0 \quad \uparrow = 0$$

this term remains

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla$$

$$\text{So } \frac{\partial}{\partial t} \left(\rho \frac{v^2}{2} \right) + \operatorname{div} \left(\rho \frac{v^2}{2} \mathbf{v} \right) = \rho \frac{D}{Dt} \left(\frac{v^2}{2} \right) \quad \checkmark$$

$$\text{Second identity} \rightarrow \frac{\partial}{\partial t} (\rho \mathbf{v}) + \operatorname{div}(\rho \mathbf{v} \mathbf{v}) = \rho \frac{D \mathbf{v}}{Dt}$$

$$\frac{\partial}{\partial t} (\rho \mathbf{v}) = \frac{\partial \rho}{\partial t} \mathbf{v} + \rho \frac{\partial \mathbf{v}}{\partial t}$$

$$\operatorname{div}(\rho \mathbf{v} \mathbf{v}) = \mathbf{v} \cdot \nabla (\rho \mathbf{v}) + \rho (\mathbf{v} \cdot \nabla) \mathbf{v}$$

$$\text{Add together} \rightarrow \frac{\partial \rho}{\partial t} \mathbf{v} + \rho \frac{\partial \mathbf{v}}{\partial t} + \mathbf{v} \cdot \nabla (\rho \mathbf{v}) + \rho (\mathbf{v} \cdot \nabla) \mathbf{v}$$

$$\rightarrow \mathbf{v} \cdot \nabla \left[\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{v}) \right] + \rho \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v}$$

$$= 0 \quad \uparrow$$

this remains

$$\frac{D \mathbf{v}}{Dt} = \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v}$$

$$\text{So } \frac{\partial}{\partial t} (\rho \mathbf{v}) + \operatorname{div}(\rho \mathbf{v} \mathbf{v}) = \rho \frac{D \mathbf{v}}{Dt} \quad \checkmark$$

#2

For fluid of constant density, $\text{div } |\nabla v|^+ = 0$

$$\frac{\partial p}{\partial t} + \text{div}(p \langle v \rangle) = 0$$

$$\text{div}(p \langle v \rangle) = p(\text{div}(\langle v \rangle)) + \langle v \rangle \cdot \nabla p$$

$$\text{Substitute} \rightarrow \frac{\partial p}{\partial t} + p(\text{div}(\langle v \rangle)) + \langle v \rangle \cdot \nabla p = 0$$

$$\text{If } p = \text{constant}, \text{ then } \frac{\partial p}{\partial t} = 0, \quad p(\text{div}(\langle v \rangle)) = 0, \quad \nabla p = 0$$

$$\text{Divided by density} \rightarrow \text{div}(\langle v \rangle) = 0$$

$$\text{In cartesian coordinates} \rightarrow \langle v \rangle = (u, v, w)$$

$$\text{So } \text{div}(\langle v \rangle) = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}$$

$$\text{So } \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

Thus, $\text{div } |\nabla v|^+ = 0$ for fluid of constant density